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Patient-independent hybrid generative-discriminative modeling for seizure detection in long-term scalp EEG

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Automated seizure detection from long-term scalp electroencephalography (EEG) remains challenging because seizure windows are sparse, channel configurations vary across patients, and clinically useful systems must maintain strict control of false alarms. This study presents a patient-independent hybrid generative-discriminative framework evaluated on the CHB-MIT cohort under leave-one-patient-out cross-validation. In each outer fold, one patient was reserved exclusively for final testing, whereas channel harmonization, representation learning, feature refinement, model training, and operating-point calibration were performed using only the remaining patients. A fold-specific convolutional β -variational autoencoder ($\beta = 1.0$) was trained on interictal windows from the training patients only. For each window, latent statistics and reconstruction error were combined with handcrafted seizure-sensitive descriptors, followed by train-only sanitization and supervised feature selection. Candidate boosted-tree classifiers, namely LightGBM, CatBoost, and XGBoost, were trained within the training pool, and operating parameters were selected on validation data only. Final performance was then computed on the held-out patient using window-level metrics with record-wise temporal postprocessing. Across the 24 held-out evaluation cases, corresponding to 23 distinct individuals in the public CHB-MIT release, the framework achieved mean accuracy of 0.9964, mean balanced accuracy of 0.9411, mean specificity of 0.9967, and mean false alarms per hour of 4.7154. Across seizure-positive held-out cases, the mean sensitivity, F1-score, ROC-AUC, and average precision were 0.5505, 0.3543, 0.9496, and 0.2990, respectively. These findings show that leakage-safe hybrid representation learning can support seizure detection while explicitly handling heterogeneous channels, severe class imbalance, and false-alarm control in long-term scalp EEG.

Keywords CHB-MIT dataset, EEG, Hybrid deep learning, Interpretable AI, Seizure detection, Variational autoencoder

Epilepsy affects more than 50 million individuals worldwide and is widely recognized as one of the most prevalent chronic neurological diseases[1,2]. Clinical diagnosis and long-term management depend critically on expert interpretation of scalp electroencephalography (EEG), which provides direct measurement of cortical electrical activity[3,4]. Manual review of prolonged EEG recordings is labor-intensive and error-prone, particularly when seizures manifest as subtle, subclinical, or artifact-obscured events[5]. From a neurophysiological perspective, seizures represent nonlinear transitions in brain dynamics[6], and both ictal and interictal EEG exhibit non-stationary characteristics, heterogeneous morphologies, and pronounced inter-patient variability. These properties pose substantial challenges for scalable and reliable automated seizure detection systems.

Classical approaches to signal processing including time domain, frequency domain, and wavelet domains have demonstrated feasibility for hand-crafted EEG feature extraction[7–9]. However, performance is significantly reduced in realistic clinical conditions with extreme seizure sparsity, variable noise profiles, and inter-patient variability. Several supervised learning pipelines assume patient-specific seizure-labeled data are available[10,11],

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which is not valid when no recorded seizures occur. As recent evaluations note[12,13], these constraints often result in poorly calibrated models, high false alarm rates per hour, and low generalizability across clinical populations.

Deep learning algorithms including long short-term memory (LSTM) networks, convolutional neural networks (CNNs), and CNN-LSTM hybrid models have shown improved performance[15,16]. Interpretable deep learning approaches have been explored to understand saliency and latent space mechanisms[17–19]. However, most deep learning approaches require substantial ictal training data, do not explicitly model normal EEG structure, and struggle with extreme class imbalance. A critical gap exists in designing a unified framework that: (a) learns a generative representation of normal EEG, (b) performs discriminative seizure detection for seizure-present patients, and (c) evaluates performance on both seizure-present and seizure-absent subjects under realistic conditions. This is essential for long-term monitoring where many patients have no seizures and false alarm suppression is critical.

To address these challenges, we develop a patient-independent hybrid generative-discriminative framework in which representation learning, feature refinement, model fitting, and calibration are confined to the training patients within each leave-one-patient-out fold. A fold-specific convolutional β-variational autoencoder (VAE) is trained on interictal windows from the training pool only, and its latent statistics and reconstruction error are combined with handcrafted seizure-sensitive descriptors. The resulting hybrid representation is refined using train-only sanitization and supervised feature selection, after which boosted-tree classifiers are fitted and calibrated on validation data drawn exclusively from the training pool. This design preserves a clean held-out-patient evaluation while retaining the advantages of unsupervised representation learning and supervised discrimination under severe class imbalance.

This study addresses three research questions: (RQ1) whether a VAE trained solely on normal EEG can maintain low false-alarm rates, including for seizure-absent patients; (RQ2) whether hybrid VAE-derived features support reliable discrimination for seizure-present patients despite extreme imbalance; and (RQ3) how anomaly behavior differs between seizure-present and seizure-absent patients in long-term monitoring. The contribution of this study does not lie in proposing a fundamentally new standalone autoencoder or classifier architecture. Rather, it lies in formulating and evaluating a leakage-safe patient-independent seizure-detection pipeline that integrates fold-specific generative representation learning, explicit channel harmonization under heterogeneous channel availability, train-only feature refinement, and validation-based operating-point calibration within a strict held-out-case LOPO design.

The principal methodological contributions of this work are as follows:

- We introduce a patient-independent hybrid seizure-detection framework that trains a fold-specific convolutional β-VAE on interictal EEG from training patients only and combines its outputs with handcrafted seizure-sensitive descriptors.
- We design the full learning pipeline to be leakage-safe within each leave-one-patient-out fold, including channel harmonization, feature sanitization, supervised feature selection, classifier fitting, and operating-point calibration without exposure to the held-out patient.
- We evaluate three boosted-tree classifiers, namely LightGBM, CatBoost, and XGBoost, under severe class imbalance using validation-based model selection and record-wise temporal postprocessing.
- We study performance across both seizure-present and seizure-absent patients and provide interpretive analyses of anomaly behavior, reconstruction error, and classifier output under a strict held-out-patient protocol.

Related work

Clinical foundations of EEG-based seizure analysis

Scalp electroencephalography is the first-line modality for diagnosis, classification, and monitoring of epilepsy[20]. Clinical manifestations are often subtle or atypical, especially in susceptible populations[21], and seizures carry significant physical and psychosocial burden[6]. From a dynamical systems perspective, seizures are transient departures from stable neural states implying nonlinear transitions in cortical dynamics[22]. These characteristics manifest in EEG as non-stationary changes in amplitude, spectral composition, and spatial organization, motivating algorithms that model variability and deviation from normal neural activity. Epilepsy is formally classified as a disease of the brain in contemporary clinical definitions, which further underscores the need for reliable monitoring and diagnostic support tools[2]. In addition, recent work on minimal-channel personalized monitoring highlights the practical importance of balancing seizure-detection performance with reduced acquisition complexity in wearable or resource-constrained settings[23].

Traditional machine learning and feature-based methods

Early automated seizure detection used manually designed features from time, frequency, and time-frequency representations. Wavelet-based statistical descriptors and nonlinear features demonstrated meaningful separation between ictal and interictal activity[7,8,16]. Classical machine learning classifiers including support vector machines, decision trees, and k-nearest neighbors showed poor generalization between patients and recording environments[10,11]. Large-scale reviews document fundamental limitations of handcrafted pipelines: sensitivity to artifacts, limited robustness to non-stationarity, and poor calibration under extreme class imbalance[12,13]. More broadly, ensemble-based anomaly-detection studies in heterogeneous settings have also shown that combining complementary learners can improve robustness under distributional variability and noisy operating conditions[14], which is conceptually relevant to long-term EEG monitoring with pronounced inter-patient heterogeneity. These challenges are critical when seizure examples are limited or absent, necessitating anomaly-oriented and patient-adaptive approaches[24].

Deep learning for seizure detection and prediction

Deep learning has advanced EEG-based seizure detection through end-to-end representation learning from raw or minimally processed signals[25,26]. LSTM networks and recurrent neural networks handle sequential dependencies with improved performance over feature-based methods[15]. Hybrid architectures using convolutional layers for spatial and spectral feature extraction and recurrent layers for temporal modeling have achieved good performance on benchmark datasets[27–29]. Extensions including residual connections, bidirectional recurrence, and contrastive learning improve temporal robustness and discriminative ability[30]. Seizure prediction employs recurrent neural networks, LSTMs, and hybrid CNN-LSTM architectures to learn pre-ictal patterns[31]. Patient-independent prediction remains difficult due to significant inter-patient variability[32,33]. Reviews emphasize that dense ictal annotations and extensive training corpora are necessary for robust performance[34–37].

As deep models have grown in complexity, interpretability has become essential for clinical adoption[38]. Saliency mapping, attribution methods, and latent space analyses elucidate seizure-related features learned by neural networks[17]. Deep models for interictal epileptiform discharge detection highlight the importance of capturing subtle patterns[18]. Reviews identify false-alarm burden, noise sensitivity, and limited interpretability as major deployment barriers[39]. Generative modeling offers a principled approach for learning normal EEG structure and quantifying deviations as anomalies[40,41].

Patient-specific and patient-independent detection

A fundamental tension exists between patient-independent generalization and patient-specific performance[42]. Patient-independent systems aim for broad applicability but often degrade for out-of-distribution subjects[43,44]. Patient-specific systems improve sensitivity but typically require explicit ictal examples and do not explicitly model normal EEG structure[45,46]. Achieving both high sensitivity and low false-alarm rates across heterogeneous patient populations remains challenging[47].

Positioning of our work

Building on prior generative and hybrid seizure-detection studies, this work does not claim novelty in the isolated use of a variational autoencoder or boosted-tree classifier. Instead, it focuses on a stricter patient-independent evaluation setting in which fold-specific representation learning, explicit channel harmonization, train-only feature refinement, model selection, and operating-point calibration are all restricted to the training cases within each leave-one-patient-out fold. The resulting framework enables seizure detection to be studied under severe class imbalance, heterogeneous channel availability, seizure-free held-out cases, and leakage-safe held-out-case evaluation. Recent CHB-MIT studies have also explored low-complexity and personalized seizure monitoring, for example by selecting a small number of channels and handcrafted features for subject-specific optimization[23]. These studies are relevant for wearable or resource-constrained settings, but their design goals differ from strict patient-independent held-out-case evaluation.

Materials and methods

System overview

Figure 1 illustrates the complete end-to-end architecture of the proposed hybrid seizure detection framework. The system comprises four tightly integrated stages: (i) fold-specific channel harmonization and window preparation, (ii) fold-specific generative representation learning on interictal windows from the training patients, (iii) hybrid feature construction with train-only sanitization and supervised feature selection, and (iv) boosted-tree classification with validation-based operating-point calibration followed by held-out-patient testing.

All experiments follow an outer leave-one-patient-out (LOPO) protocol with 24 independent evaluation folds. In each outer fold, one patient is reserved for final testing and the remaining patients form the training pool. The training pool is further divided into inner training and validation subsets, preferably using grouped patient splitting, with a stratified fallback only when required to preserve both classes. The held-out patient is not used for VAE training, feature selection, model fitting, calibration, or any other optimization step. Because channel layouts differed across subjects, no fixed universal electrode set or new rereferencing scheme was imposed during this stage of the analysis.

Dataset and cohort

All experiments were conducted using the publicly available CHB-MIT pediatric scalp EEG dataset[48–50], sampled at 256 Hz. In the public release used here, the benchmark configuration comprises 24 evaluation cases corresponding to 23 distinct individuals, because case chb21 is a later recording from the same individual as chb01, whereas chb24 was added separately in a later release. In this study, all 24 available evaluation cases were retained in order to preserve the standard benchmark configuration used in the prepared cohort. Each evaluation case was categorized as seizure-positive or seizure-free according to whether any retained evaluation window carried a seizure label. Seizure annotations were not filtered according to seizure duration.

Continuous EEG recordings were segmented into partially overlapping 5-second windows (1280 samples at 256 Hz) with 50% overlap. Per-window metadata retained at least the recording file, segment start time, and case identity. The same channel set was not assumed to be consistently available across all cases. Instead, within each outer LOPO fold, channel names were first normalized on the training cases only, after which channels were prioritized by training-case presence frequency to define a fold-specific canonical layout. Consequently, the exact channel identities were fold dependent, and the canonical set size ranged from 16 to 23 channels across folds. Accordingly, the framework does not rely on a single universal 32-channel specification, but instead uses the subset of channels that can be harmonized consistently within each fold from the available training cases. Held-out windows were then aligned to this fold-specific layout by channel name rather than raw column index, and missing or unavailable channels were represented explicitly through a binary channel mask.

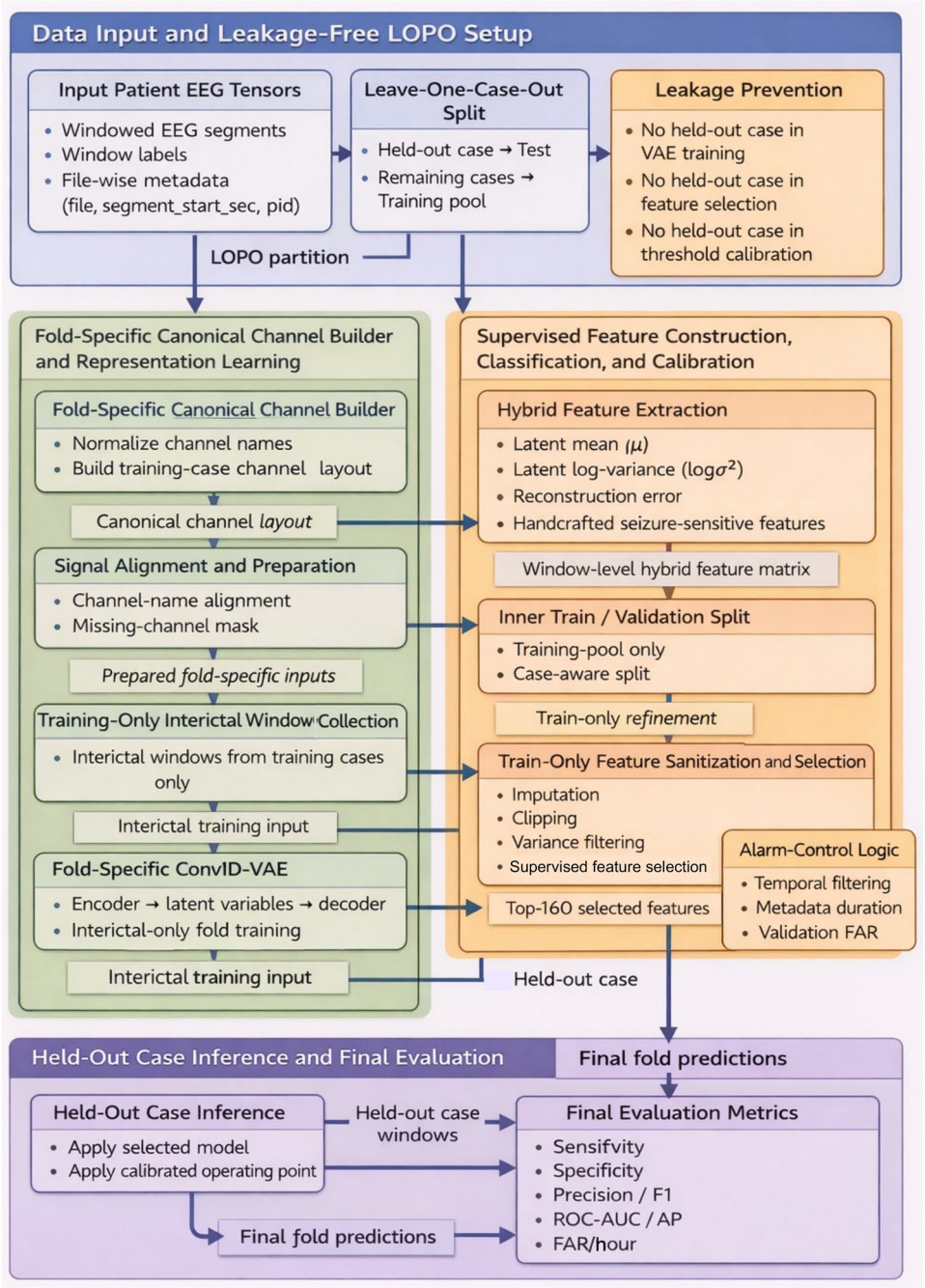


Fig. 1. Overview of the seizure-detection pipeline under the outer leave-one-patient-out design. Within each fold, the training patients undergo channel harmonization, fold-specific convolutional β-VAE training, hybrid feature construction, train-only feature refinement, boosted-tree model fitting, and validation-based operating-point calibration. The held-out patient is used only for final inference and evaluation.

The runner operated on already prepared EEG tensors and did not introduce an additional rereferencing or montage-conversion step such as bipolar, monopolar, common-average, or global-average referencing. Therefore, no new reference strategy is claimed within this stage of the pipeline. False alarms per hour were computed as the total number of false-positive windows divided by the total evaluated EEG duration in hours for the held-out case, where duration was derived from file-level metadata and actual window start times rather than from a fixed denominator. This avoided underestimation of alarm burden under overlapping-window evaluation and preserved correct record-wise duration accounting.

Artifact handling

The reported seizure-detection pipeline operated directly on the prepared EEG tensors provided to the LOPO framework and did not apply a separate artifact-gating network before variational autoencoder training or

classifier evaluation. Robustness to noisy or atypical windows was addressed through fold-specific representation learning on training-patient interictal data, train-only feature sanitization, supervised feature selection, and validation-based operating-point calibration. Accordingly, artifact handling in the main pipeline should be understood as being implicit within the representation-learning, feature-refinement, and calibration stages rather than as a distinct preclassification rejection module. No separate quantitative artifact-rejection analysis is reported for the main results, and the present study therefore does not attribute performance gains to an independently validated artifact-filtering module.

Variational autoencoder architecture

Within each outer LOPO fold, a convolutional β-VAE with $\beta = 1.0$ was trained using interictal windows from the training patients only, producing a fold-specific generative model of non-seizure EEG. The model captures the structure and variability of clean interictal activity within the training pool, while the held-out patient remains completely unseen during representation learning. The encoder consisted of four Conv1D layers (32, 64, 128, 256 filters) with batch normalization, LeakyReLU ($\alpha = 0.2$), and dropout ($p = 0.1$). The encoder outputs latent mean $\boldsymbol{\mu} \in \mathbb{R}^{64}$ and log-variance $\log \boldsymbol{\sigma}^2 \in \mathbb{R}^{64}$. Latent sampling uses the reparameterization trick:

$$\mathrm{z} = \boldsymbol{\mu} + \exp\left(\frac{1}{2}\log \boldsymbol{\sigma}^2\right) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathrm{I}) \tag{1}$$

The decoder uses ConvTranspose1D operations to reconstruct window $\hat{x}$ from latent vector z. Reconstruction quality is measured as mean-squared error (MSE) between input x and reconstruction $\hat{x}$, producing scalar reconstruction error r.

The training objective combines reconstruction fidelity and latent regularization:

$$\mathcal{L} = \mathrm{MSE}(x, \hat{x}) + \beta D_{\mathrm{KL}}(q(\mathrm{z}|x) \| \mathcal{N}(0, \mathrm{I})) \tag{2}$$

With $\beta = 1.0$, reconstruction and KL divergence terms are balanced. The KL divergence enforces smoothness in the latent space by encouraging the posterior toward a unit Gaussian prior.

Latent feature extraction

For each EEG window, the encoder outputs latent mean $\boldsymbol{\mu} \in \mathbb{R}^{64}$ and latent log-variance $\log \boldsymbol{\sigma}^2 \in \mathbb{R}^{64}$, and reconstruction error $r \in \mathbb{R}$ is computed from the input and reconstruction. These VAE-derived quantities are combined with handcrafted seizure-sensitive descriptors computed from the aligned raw signals, including log-scaled amplitude and variability measures, relative band powers in the delta, theta, alpha, beta, and gamma ranges, and present-channel summary statistics. Sampled latent vectors are not used as classifier inputs.

After feature construction, NaN and Inf values are replaced using training medians, extreme values are clipped using training quantiles, near-constant features are removed, and a train-only LightGBM selector reduces the representation to the top 160 features used for classification. Therefore, the classifier does not operate on a fixed 193-dimensional latent vector, but on a train-refined hybrid representation whose final dimensionality is determined after fold-specific feature construction and train-only selection.

Hybrid classifier training

Classification was performed within each outer LOPO fold using only features derived from the training patients. The training pool was further divided into inner training and validation subsets, preferably with grouped patient splitting and a stratified fallback only when required to preserve both classes. Three boosted-tree models were evaluated: LightGBM, CatBoost, and XGBoost. To address severe class imbalance, all seizure windows were retained, non-seizure windows were capped at a 20:1 ratio, and positive-class sample weighting was applied with an upper limit of 10.0. This strategy was intended to avoid artificial 1:1 balancing while still limiting the dominance of non-seizure windows during model fitting. Model selection was based on validation performance only.

Threshold and alarm calibration

Classifier outputs represent seizure probabilities in [0, 1]. Operating parameters were selected exclusively within the training pool. For each outer LOPO fold, candidate settings were evaluated on validation data only by jointly searching decision threshold $\theta \in \{0.05, 0.06, \ldots, 0.99\}$, temporal smoothing window $w \in \{1, 3, 5\}$, and minimum consecutive-positive run length $m \in \{1, 2\}$. Candidate settings were further constrained by pooled validation false-alarm-rate caps $\{0.5, 1, 2, 5, 10, 20\}$ and by an additional guardrail limiting false alarms per hour on negative-only validation patients to at most 10.0. The final operating point was selected using validation performance only and was then applied without modification to the held-out patient.

Final evaluation

Final detection metrics were computed for the held-out patient after record-wise temporal postprocessing applied separately within each EEG file. Sensitivity, specificity, precision, and F1-score were defined as:

$$\mathrm{Sensitivity} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \quad \mathrm{Specificity} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{3}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{F1} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{4}$$

Balanced accuracy was defined as:

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \tag{5}$$

False alarms per hour were computed using metadata-derived recording duration:

$$\text{FAR/h} = \frac{\text{FP}}{H_{\text{test}}} \tag{6}$$

where H_{test} denotes the total evaluated recording duration in hours for the held-out patient. This quantity was derived from file-level metadata and actual window start times within the held-out recordings, so the alarm-rate calculation reflects true monitored duration rather than an assumed constant number of windows per hour. ROC-AUC and average precision were computed from the predicted seizure probabilities. This evaluation was repeated for all 24 LOPO folds to obtain the final patient-level performance profile.

Experimental setup

All experiments were conducted on a workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), Intel Xeon 2.7 GHz CPU, 64 GB RAM, and running Python 3.9 with PyTorch 2.1, scikit-learn, LightGBM, CatBoost, and XGBoost libraries. Random seeds across NumPy, PyTorch, and classifier libraries were fixed to improve reproducibility.

Within each outer LOPO fold, the convolutional β-VAE was trained on interictal windows from the training patients only using AdamW optimizer (learning rate 10^{-4}, weight decay 10^{-5}), batch size 64, early stopping, and loss $\text{MSE}(x, \hat{x}) + \beta D_{\text{KL}}$. Hybrid features were then constructed for the training, validation, and held-out test windows of that fold and refined using train-only sanitization and supervised feature selection. Candidate boosted-tree classifiers were trained on the inner training subset, evaluated on validation data, and calibrated using validation-only operating-point search over threshold, temporal smoothing, and minimum-run settings. After fold-wise model selection, final performance was computed on the held-out patient and aggregated across all 24 LOPO folds. Fold-specific channel harmonization metadata, including the canonical channel count and alignment summary, were recorded for each LOPO fold to support transparent handling of heterogeneous channel availability.

Results

All results in this section were obtained under the leave-one-patient-out (LOPO) evaluation procedure described in Section "Materials and methods". Within each fold, channel harmonization, fold-specific representation learning, feature sanitization, feature selection, model selection, and operating-point calibration were restricted to the training cases, whereas the held-out case was reserved exclusively for final evaluation. Accordingly, all performance values reported below reflect patient-independent held-out testing.

Dataset composition and preprocessing outcomes

Across the 24 held-out LOPO folds, the final evaluation covered 1,414,710 windows, of which 876 were seizure windows and 1,413,834 were non-seizure windows. The evaluated cohort consisted of 24 held-out cases from 23 distinct individuals. Six held-out cases (chb01, chb02, chb03, chb04, chb05, and chb24) contained seizure-labeled windows, whereas the remaining 18 held-out cases were seizure-free. Windows were represented as 5-second segments with 50% overlap and were aligned within each fold by channel name to a training-derived canonical channel layout rather than to a fixed 32-channel tensor. For each fold, a fold-specific β-VAE trained on interictal windows from the training cases generated latent statistics and reconstruction error, which were combined with handcrafted seizure-sensitive descriptors and reduced to 160 selected features through train-only sanitization and LightGBM-based feature selection. Seizure windows constituted approximately 0.062% of all evaluated windows, underscoring the severe class imbalance of the task. Table 1 summarizes the dataset composition, preprocessing decisions, and fold-specific input handling under the corrected LOPO evaluation protocol.

Overall hybrid detection performance

Table 2 summarizes aggregate performance under the LOPO protocol. Across all 24 held-out evaluation cases, the framework achieved mean accuracy of 0.9964, mean balanced accuracy of 0.9411, mean specificity of 0.9967, and mean false alarms per hour (FAR/h) of 4.7154. Because 18 of the 24 held-out cases were seizure-free, the all-case averages are dominated by non-seizure discrimination. For seizure-detection ability, the six seizure-positive folds are more informative. On this subset, the mean sensitivity, F1-score, ROC-AUC, and average precision (AP) were 0.5505, 0.3543, 0.9496, and 0.2990, respectively. These results indicate that the pipeline preserved strong non-seizure rejection while achieving moderate seizure capture under patient-independent testing. All FAR/h values reported in this section were computed using the metadata-derived held-out recording duration for each case.

Patient-specific detection performance

Table 3 reports patient-level performance and the selected best model for each held-out fold. CatBoost was selected in 17 folds, LightGBM in 5 folds, and XGBoost in 2 folds, indicating that the final operating configuration remained patient dependent even under a common leakage-safe pipeline. Among seizure-positive held-out cases, sensitivity ranged from 0.3839 (chb05) to 0.8500 (chb01), and F1-score ranged from 0.0286 (chb04) to

Component	Description
Dataset	24 CHB-MIT cases
Outer evaluation	24 LOPO folds
Window length	5 seconds
Overlap	50%
Sampling rate	256 Hz
Total evaluated windows	1,414,710
Seizure windows	876
Non-seizure windows	1,413,834
Held-out seizure-positive cases	6 (chb01, chb02, chb03, chb04, chb05, chb24)
Held-out seizure-free cases	18
Distinct individuals	23
Channel availability	Not identical across subjects
Canonical channel handling	Fold-dependent training-derived layout (16–23 channels), no fixed 32-channel tensor
Missing-channel handling	Channel-name alignment with binary channel mask
Montage/reference in runner	No additional rereferencing introduced at runner stage
Feature representation	Latent $\boldsymbol{\mu}$, latent log $\boldsymbol{\sigma}^2$, reconstruction error, handcrafted descriptors
Final classifier input	Top 160 selected features
Artifact handling in main run	No separate artifact-gating model applied
Evaluation	LOPO with validation-only model selection and calibration

Table 1. Dataset and preprocessing summary under the LOPO evaluation protocol.

Summary item	Value
Held-out evaluation cases	24
Distinct individuals	23
Held-out seizure-positive cases	6
Held-out seizure-free cases	18
Mean accuracy (all cases)	0.9964
Mean balanced accuracy (all cases)	0.9411
Mean specificity (all cases)	0.9967
Mean FAR/h (all cases)	4.7154
Mean sensitivity (seizure-positive cases)	0.5505
Mean F1-score (seizure-positive cases)	0.3543
Mean ROC-AUC (seizure-positive cases)	0.9496
Mean AP (seizure-positive cases)	0.2990

Table 2. Aggregate performance summary under LOPO evaluation.

0.6860 (chb24). The most difficult positive fold was chb04, where sensitivity of 0.5156 was accompanied by precision of 0.0147 and FAR/h of 14.1419, indicating substantial false-positive activity despite partial seizure capture. Among seizure-free cases, specificity ranged from 0.9849 to 1.0000, and seven folds achieved zero false alarms. Overall, the patient-level results show that strong non-seizure rejection can be maintained consistently across the cohort, whereas seizure capture remains substantially more variable across subjects.

Patient-wise performance landscape

Figure 2 summarizes the patient-level LOPO results. The left panel shows sensitivity, specificity, F1-score, ROC-AUC, and AP, whereas the right panel shows FAR/h for the same held-out cases. Seizure-positive and seizure-free cases exhibit distinctly different performance regimes. Among seizure-positive cases, ROC-AUC remained high in all six folds, ranging from 0.8850 to 0.9828, indicating that the learned scores preserved strong ranking ability even when operating-point performance varied. However, AP and F1-score were more heterogeneous, with the strongest combined profile observed for chb01 and chb24 and the weakest for chb04. In seizure-free cases, sensitivity, ROC-AUC, and AP are not defined because no seizure windows were present in the held-out case. For those folds, specificity and FAR/h provide the relevant interpretation. The figure therefore highlights two central properties of the pipeline: consistently strong non-seizure rejection across the full cohort and subject-dependent seizure capture among the positive cases.

Patient	Best model	Sens.	Spec.	Prec.	F1	ROC-AUC	AP	FAR/h
Seizure-positive held-out cases								
chb01	CatBoost	0.8500	0.9967	0.4448	0.5840	0.9828	0.7585	4.7100
chb02	CatBoost	0.4595	0.9969	0.0971	0.1604	0.9753	0.0515	4.4802
chb03	CatBoost	0.4110	0.9959	0.2326	0.2971	0.9805	0.2048	5.8156
chb04	CatBoost	0.5156	0.9902	0.0147	0.0286	0.9002	0.0316	14.1419
chb05	XGBoost	0.3839	0.9972	0.3568	0.3699	0.9739	0.2479	3.9741
chb24	LightGBM	0.6827	0.9979	0.6893	0.6860	0.8850	0.4995	3.0053
Seizure-free held-out cases								
chb06	CatBoost	–	0.9944	0.0000	0.0000	–	–	8.0469
chb07	CatBoost	–	0.9849	0.0000	0.0000	–	–	21.8042
chb08	CatBoost	–	0.9943	0.0000	0.0000	–	–	8.1974
chb09	CatBoost	–	0.9889	0.0000	0.0000	–	–	16.0013
chb10	CatBoost	–	0.9979	0.0000	0.0000	–	–	3.0786
chb11	CatBoost	–	0.9950	0.0000	0.0000	–	–	7.1566
chb12	CatBoost	–	1.0000	0.0000	0.0000	–	–	0.0000
chb13	XGBoost	–	1.0000	0.0000	0.0000	–	–	0.0000
chb14	LightGBM	–	0.9999	0.0000	0.0000	–	–	0.1923
chb15	CatBoost	–	1.0000	0.0000	0.0000	–	–	0.0000
chb16	CatBoost	–	1.0000	0.0000	0.0000	–	–	0.0000
chb17	CatBoost	–	1.0000	0.0000	0.0000	–	–	0.0000
chb18	LightGBM	–	1.0000	0.0000	0.0000	–	–	0.0000
chb19	CatBoost	–	1.0000	0.0000	0.0000	–	–	0.0000
chb20	CatBoost	–	0.9993	0.0000	0.0000	–	–	0.9783
chb21	LightGBM	–	0.9996	0.0000	0.0000	–	–	0.5178
chb22	CatBoost	–	0.9975	0.0000	0.0000	–	–	3.6126
chb23	LightGBM	–	0.9948	0.0000	0.0000	–	–	7.4559

Table 3. Patient-level performance under LOPO evaluation.

Patient-wise prediction profile

Figure 3 reports the patient-wise TP/FP/FN/TN profile matrix. True positives were concentrated entirely within the six seizure-positive held-out cases, with chb01 and chb24 producing the largest TP counts (153 and 142, respectively). The corresponding FN counts indicate that seizure capture remained incomplete in every positive fold, particularly for chb05 and chb24. The main contrast among seizure-free folds lies in their FP counts. Several seizure-free cases showed no false positives at all (chb12, chb13, chb15, chb16, chb17, chb18, and chb19), whereas others produced nontrivial FP burdens, most prominently chb07 and chb09. This patient-wise count view complements Table 3 by showing that the principal difficulty of the task is not only seizure rarity, but also the highly uneven distribution of false alarms across subjects.

Seizure-positive case analysis

Figure 4 focuses on the six seizure-positive held-out cases and reports sensitivity, ROC-AUC, AP, and FAR/h. This subset is clinically more informative than the all-case average because seizure-free folds can inflate global accuracy and specificity. The results reveal several distinct positive-case regimes. chb01 showed the strongest seizure capture, with sensitivity of 0.8500, ROC-AUC of 0.9828, AP of 0.7585, and FAR/h of 4.7100. chb24 showed the most balanced joint profile, with sensitivity of 0.6827, F1-score of 0.6860, AP of 0.4995, and the lowest FAR/h among the seizure-positive folds. By contrast, chb02 and chb03 retained high ROC-AUC values despite weak AP, indicating that score ranking remained strong but the selected operating point captured relatively few positive windows under severe imbalance. The most challenging case was chb04, where sensitivity of 0.5156 co-occurred with AP of only 0.0316 and FAR/h of 14.1419, reflecting substantial overlap between seizure-like and non-seizure transients in the held-out recording. Together, these positive-case results indicate that the pipeline can recover meaningful seizure discrimination across multiple patients, but that the final operating characteristics remain strongly patient dependent.

False-alarm burden in seizure-free cases

Figure 5 isolates FAR/h for the 18 seizure-free held-out cases. The distribution was highly skewed. The mean FAR/h across seizure-free cases was 4.28, whereas the median was 0.75, indicating that a relatively small number of difficult cases dominated the average false-alarm burden. chb07 and chb09 showed the highest FAR/h values (21.8042 and 16.0013, respectively), followed by chb08, chb06, chb23, and chb11. In contrast, seven seizure-free folds achieved zero false alarms, and three additional cases (chb14, chb20, and chb21) remained below 1 FAR/h. These findings show that the framework can suppress false alarms effectively in many seizure-free cases, but

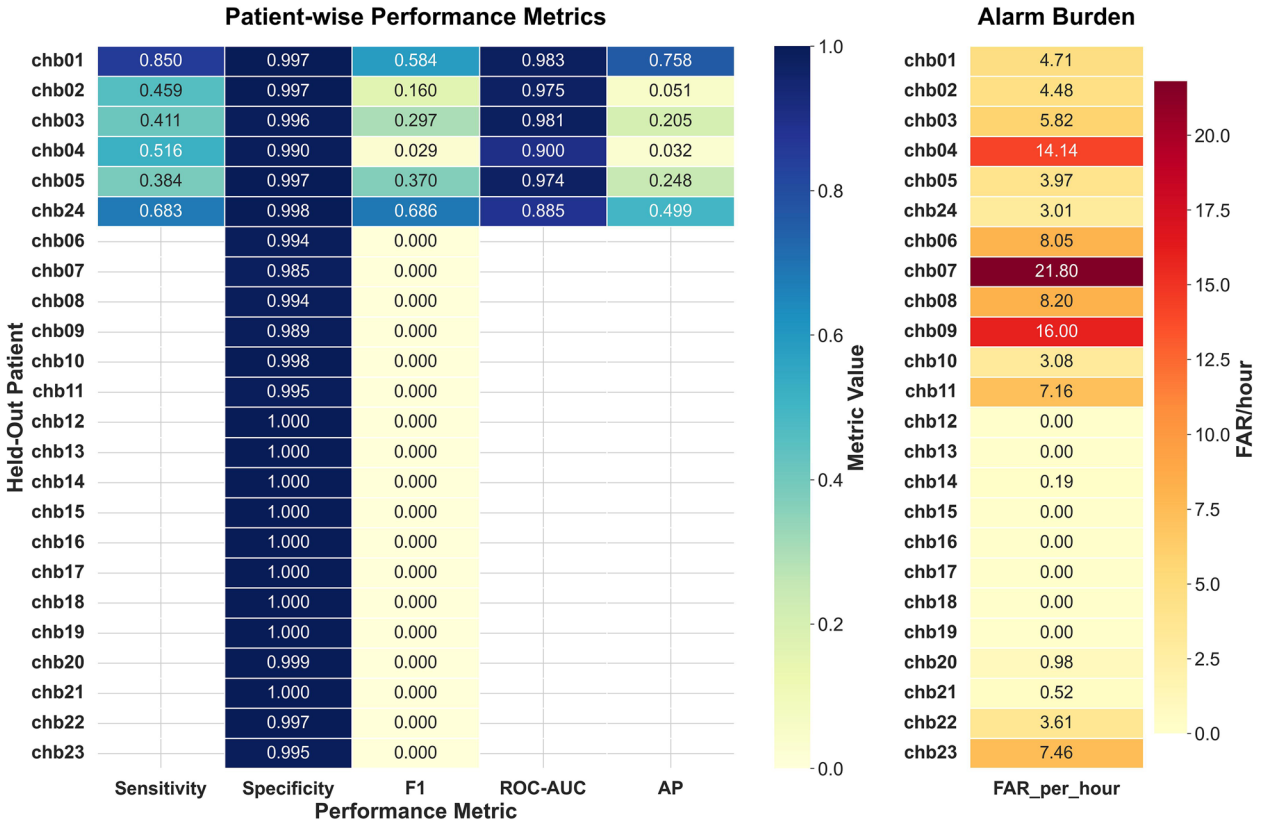


Fig. 2. Patient-wise performance heatmap under LOPO evaluation. The left panel summarizes sensitivity, specificity, F1-score, ROC-AUC, and average precision for each held-out case. Sensitivity, ROC-AUC, and AP are shown only for seizure-positive held-out cases; these quantities are not defined for seizure-free cases and are therefore omitted. The right panel reports the corresponding FAR/h values for all held-out cases.

residual paroxysmal or atypical non-seizure activity continues to challenge uniform alarm control across the full cohort.

Discussion

Interpretation under leakage-safe held-out evaluation

The present results show that the proposed framework can maintain strong non-seizure discrimination under a strict held-out-patient protocol while providing more moderate seizure sensitivity in seizure-positive cases. This distinction is important because the full pipeline, including channel harmonization, representation learning, feature sanitization, feature selection, model selection, and operating-point calibration, was restricted to the training cases in each fold. Consequently, the results should be interpreted as patient-independent held-out performance. Under this setting, the principal strength of the framework lies in its ability to preserve very high specificity across the cohort, whereas seizure capture remains more heterogeneous across positive cases.

Latent anomaly modeling and hybrid features

The fold-specific β-VAE was trained only on interictal windows from the training cases. Consequently, elevated reconstruction error in a held-out case reflects deviation from a training-derived interictal reference distribution rather than from a globally learned subject-mixed baseline. In the present framework, reconstruction error is not interpreted as a direct biological biomarker by itself. Instead, it serves as one component of a broader hybrid representation that helps identify windows whose signal characteristics differ from the interictal patterns observed during training.

The final decision model does not rely on reconstruction error alone. Each held-out window is represented using latent mean, latent log-variance, reconstruction error, and handcrafted seizure-sensitive descriptors, followed by train-only sanitization and feature selection before classification. This design is important because ambiguous non-seizure transients may also yield elevated reconstruction error. Seizure likelihood is therefore determined from the joint behavior of latent statistics, reconstruction error, and handcrafted descriptors through the downstream classifier and validation-based operating-point calibration.

This hybrid formulation is consistent with the patient-wise results reported in Section "Experimental setup". The framework preserved strong ranking performance in seizure-positive cases, as reflected by high ROC-AUC values, while the final operating characteristics remained sensitive to patient-specific overlap between seizure and non-seizure dynamics. In this sense, the latent anomaly component contributes useful deviation-sensitive

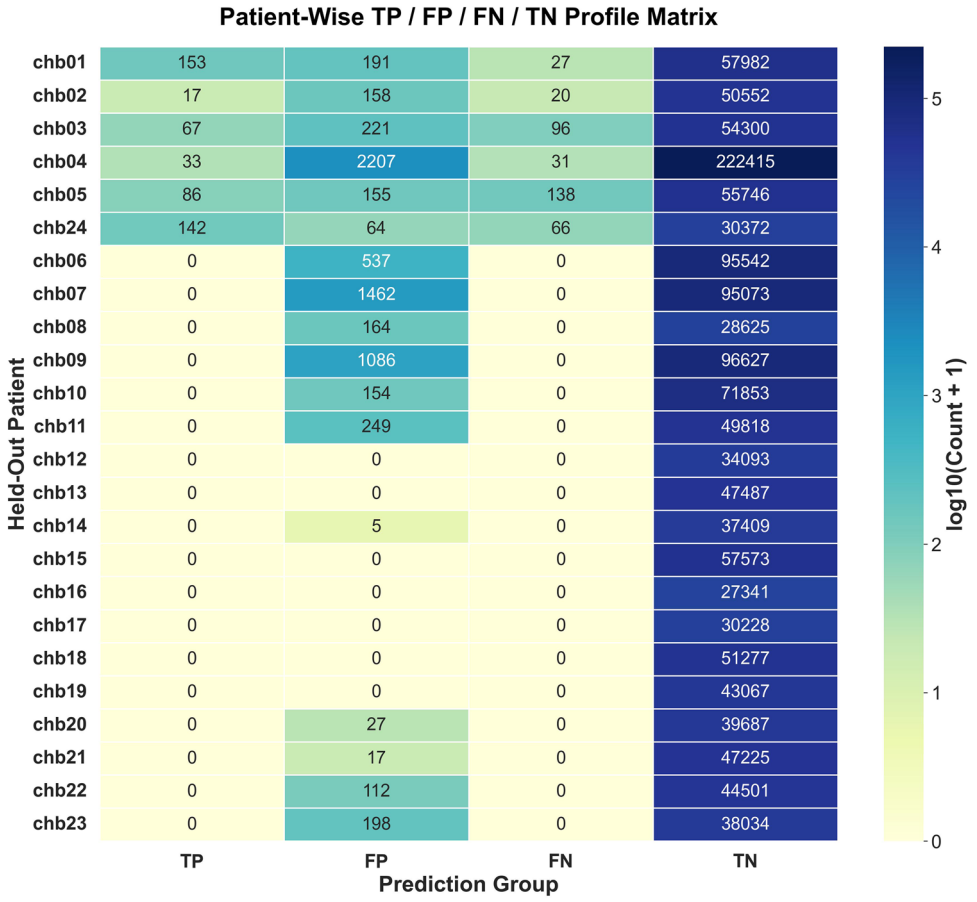


Fig. 3. Patient-wise TP/FP/FN/TN profile matrix under LOPO evaluation. Cell annotations denote raw counts, whereas the color scale reflects $\log_{10}(\text{count} + 1)$ to preserve visibility across the large dynamic range of segment counts.

information, but its value emerges primarily when interpreted jointly with the remaining hybrid features rather than in isolation.

Physiological interpretability of the hybrid representation

The physiological interpretability of the framework comes primarily from the handcrafted seizure-sensitive descriptors that accompany the learned latent representation. These descriptors include log-scaled amplitude measures, variability-related statistics, and relative band-power features in the delta, theta, alpha, beta, and gamma ranges. In practical EEG terms, these feature families reflect signal properties that often change during ictal activity, including altered rhythmic organization, redistribution of spectral energy, and increased amplitude instability.

Accordingly, the hybrid representation is not limited to a generic statement that seizure windows merely deviate from normal interictal behavior. The downstream classifier receives both anomaly-related information from the fold-specific β-VAE and physiologically interpretable signal descriptors derived directly from the aligned EEG windows. The value of this design is that seizure likelihood is informed not only by deviation from an interictal reference distribution, but also by measurable changes in amplitude, variability, and spectral composition that can be discussed in conventional EEG terms. Reconstruction error should therefore be interpreted as a nonspecific anomaly indicator within a broader discriminative representation rather than as a direct biological marker on its own.

This interpretation also helps explain why patient-wise performance remains heterogeneous across seizure-positive cases. In some held-out patients, seizure-related changes in amplitude and spectral organization appear sufficiently distinct to support favorable joint performance, whereas in more difficult cases the feature space still shows appreciable overlap between ictal and seizure-like non-ictal activity. Thus, the hybrid representation improves interpretability and discrimination simultaneously, but does not eliminate the inherent physiological variability of long-term scalp EEG.

Performance on seizure-positive cases

The seizure-positive results demonstrate that the framework can recover meaningful seizure discrimination under extreme imbalance, but not uniformly across all positive patients. The positive-case panel in Figure 4

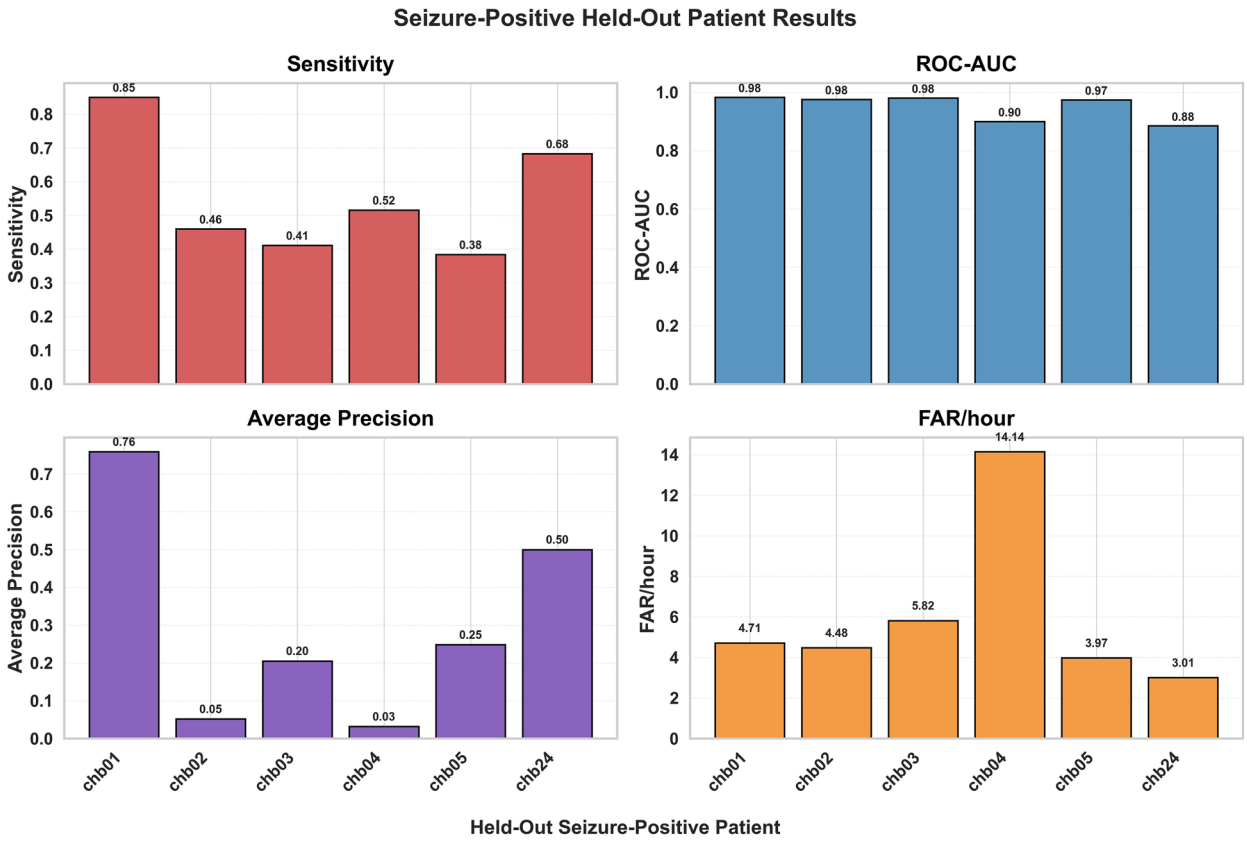


Fig. 4. Performance on seizure-positive held-out cases. The four panels report sensitivity, ROC-AUC, average precision, and FAR/h for chb01, chb02, chb03, chb04, chb05, and chb24 under LOPO evaluation.

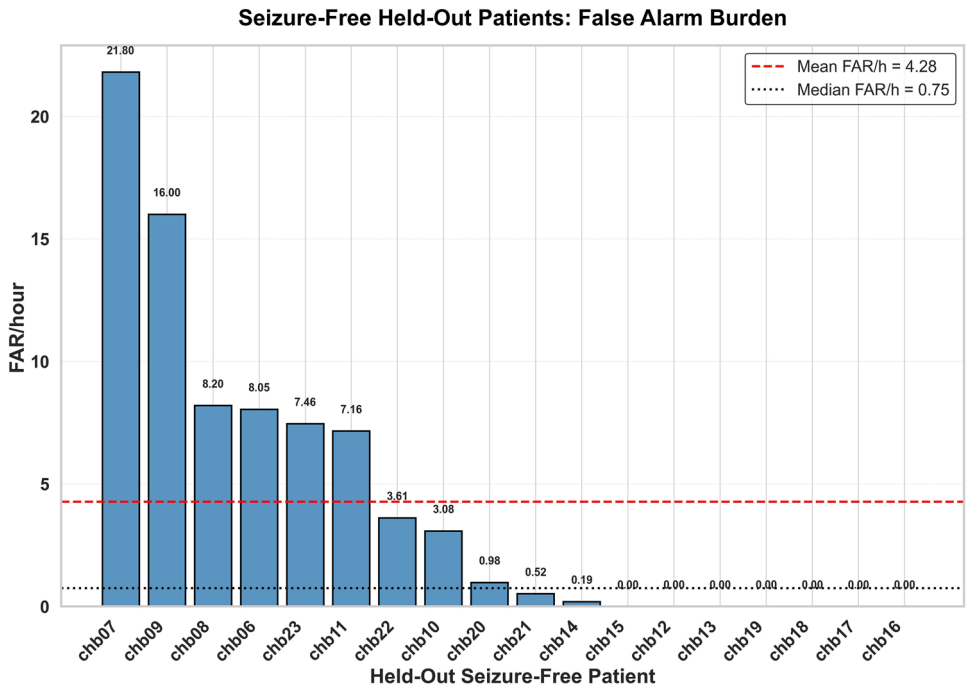


Fig. 5. False-alarm burden across seizure-free held-out cases. The bars report FAR/h for each seizure-free fold under LOPO evaluation. The dashed and dotted lines denote the mean and median FAR/h across seizure-free cases, respectively.

Study	Evaluation setting	Channel/feature strategy	Key reported outcome
Peng et al. (2021)[23]	Personalized seizure monitoring on 23 subjects	Up to 3 channels and no more than 7 selected features	Average F1-score of 88%
Ali et al. (2024)[51]	Generalized cross-subject event detection on continuous EEG	22 channels, 5-second non-overlapping windows, Random Forest classifier	Event sensitivity of 72.63% (subject-wise 5-fold) and 75.34% (leave-one-out)
This study	Patient-independent LOPO over 24 evaluation cases (23 distinct individuals)	Fold-dependent canonical layout (16–23 channels), hybrid latent + handcrafted features, top-160 selected features	Mean specificity of 0.9967 and mean FAR/h of 4.7154 across all held-out cases; mean sensitivity of 0.5505 and mean ROC-AUC of 0.9496 across seizure-positive held-out cases

Table 4. Comparison with representative CHB-MIT studies.

shows that chb01 and chb24 achieved the most favorable overall operating profiles, whereas chb02 and chb03 retained high ROC-AUC values but low AP values, indicating that ranking quality remained strong even when the selected threshold captured relatively few positive windows. The chb04 fold was particularly difficult because it combined moderate sensitivity with very low precision and the highest FAR/h among the seizure-positive cases. This behavior suggests that the score space retained useful seizure-related ordering information, but that seizure-like non-ictal transients remained difficult to suppress at the final operating point in some patients. The patient-wise confusion matrix in Figure 3 reinforces this interpretation by showing that seizure capture and false-positive burden varied substantially even within the seizure-positive subset.

False-alarm burden and conservative operating behavior

The seizure-free FAR/h analysis in Figure 5 indicates that false-alarm control improved substantially in many held-out cases, but was not uniform across the cohort. Seven seizure-free folds produced zero false alarms and three additional folds remained below 1 FAR/h, demonstrating that the validation-only calibration strategy can yield highly conservative behavior in a substantial portion of the dataset. At the same time, cases such as chb07 and chb09 still showed appreciable alarm burden. This pattern indicates that the current operating regime is conservative overall and favors non-seizure rejection, yet remains vulnerable to patient-specific EEG patterns that are seizure-like in the learned feature space. From a practical monitoring perspective, this trade-off is understandable, but it also highlights that alarm burden remains a central limitation of the present system rather than a fully solved problem.

Heterogeneity, channel harmonization, and fold-wise model selection

An important strength of the proposed framework is that it explicitly accommodates channel heterogeneity. Instead of imposing a fixed 32-channel tensor, each fold constructs a canonical channel layout from the training cases only, aligns windows by channel name, and uses a binary missing-channel mask to preserve information about absent channels. This design helps avoid conflating genuine physiological differences with arbitrary channel-order artifacts. The fact that CatBoost, LightGBM, and XGBoost were all selected across different folds further indicates that no single classifier dominated the cohort universally. Rather, fold-wise model selection appears to reflect meaningful patient-level heterogeneity in both channel availability and signal morphology.

Addressing the research questions

RQ1: Can a VAE trained on interictal EEG support low false-alarm operation?
The results support a qualified affirmative answer. Across all 24 held-out cases, mean specificity was 0.9967 and mean FAR/h was 4.7154. Among seizure-free cases, several folds achieved zero false alarms and the median FAR/h remained low. However, the seizure-free FAR/h distribution was highly skewed, with a small number of difficult cases accounting for much of the residual alarm burden. Therefore, the fold-specific VAE contributes meaningfully to non-seizure discrimination, but low false-alarm operation is not uniformly guaranteed across all patients.

RQ2: Do hybrid features support seizure discrimination under severe imbalance?
The results again support a qualified affirmative answer. Across seizure-positive held-out cases, mean sensitivity was 0.5505, mean F1-score was 0.3543, mean ROC-AUC was 0.9496, and mean AP was 0.2990. These values show that the hybrid representation preserved substantial seizure-discriminative information even under extreme imbalance and strict held-out evaluation. Nevertheless, the seizure-positive folds also showed marked heterogeneity, ranging from strong detection behavior in chb01 and chb24 to much weaker AP and precision in chb02, chb03, and especially chb04. Thus, the hybrid representation is clearly useful, but its final decision behavior remains patient dependent.

RQ3: How does the framework behave across seizure-positive and seizure-free cases?
The results show a clear distinction between the two groups. Seizure-positive cases exhibit nonzero TP and FN counts together with substantially more variable sensitivity, AP, and FAR/h, whereas seizure-free cases are characterized by uniformly undefined seizure-capture metrics, near-perfect specificity, and highly variable but often low FAR/h. In other words, the principal challenge in seizure-positive cases is balancing seizure capture against false positives, whereas the principal challenge in seizure-free cases is suppressing a relatively small number of difficult false-alarm subjects. This contrast is shown directly by the patient-wise performance heatmap, the TP/FP/FN/TN profile matrix, and the seizure-free FAR/h plot.

Comparison with representative CHB-MIT studies

Because CHB-MIT is a widely used benchmark, the present results should be interpreted alongside representative prior studies that pursued different operating assumptions. Peng *et al.* presented a personalized low-complexity seizure-monitoring framework on CHB-MIT and reported an average F1-score of 88% using no more than seven selected features from at most three channels[23]. However, that study follows a personalized optimization strategy and is therefore not directly comparable to a strict held-out-case patient-independent protocol. Ali *et al.* investigated generalized cross-subject seizure event detection on continuous CHB-MIT EEG using 22 channels, 5-second non-overlapping windows, and a Random Forest classifier, and reported event sensitivities of 72.63% and 75.34% for subject-wise 5-fold and leave-one-out analyses[51]. That study addressed a more event-oriented detection scenario, whereas the present work remains a window-level held-out-case analysis with record-wise temporal postprocessing.

Taken together, these comparisons show that published CHB-MIT results are strongly shaped by study design choices, including whether the task is personalized or patient-independent, event-level or window-level, low-complexity channel selection or fold-wise channel harmonization, and whether seizure-free held-out cases are explicitly retained. The present study should therefore be interpreted as a leakage-safe patient-independent hybrid benchmark emphasizing strict held-out-case evaluation, explicit handling of heterogeneous channel availability, train-only feature refinement, and validation-based operating-point calibration rather than as a direct substitute for personalized monitoring systems or formal event-level detectors.

Summary and implications

Taken together, the results support the feasibility of patient-independent hybrid seizure detection in long-term scalp EEG when the entire pipeline is evaluated under a LOPO design. The framework handles heterogeneous channel availability, severe class imbalance, and validation-based operating-point calibration while maintaining very strong specificity. At the same time, seizure sensitivity remains variable across seizure-positive patients, and alarm burden remains non-negligible in a subset of seizure-free cases. The present results therefore position the method as a rigorous leakage-safe patient-independent benchmark rather than as a completed clinical solution. More broadly, the study shows that hybrid latent and handcrafted representations can support meaningful seizure discrimination under strict held-out evaluation, but that stronger alarm control and more uniform seizure capture remain important directions for future work.

Limitations and future work

Limitations

The framework was evaluated only on the CHB-MIT dataset[48,49], which comprises pediatric scalp EEG recorded in a controlled hospital environment. Although the LOPO design is strict and covers all 24 subjects, generalizability to adult, ambulatory, and multi-center EEG settings has not yet been established.

Performance among seizure-positive patients remained uneven. Sensitivity ranged from 0.3839 to 0.8500, and precision ranged from 0.0147 to 0.6893, indicating that some seizure morphologies and some false-positive regimes remain difficult for the current representation and calibration strategy. In particular, chb04 achieved sensitivity above 0.5 but with very low precision because false alarms remained frequent. This variability shows that clinically useful seizure capture has not yet been reached uniformly across patients.

The current study is still a window-level detection analysis with record-wise temporal postprocessing. It does not yet report event-level sensitivity, onset delay, or alarm-level burden, which are important for clinical interpretation.

No separately validated artifact-gating stage was included in the main reported experiments. As a result, the robustness of the framework to heavily contaminated recordings remains an open question. In addition, the study does not yet include a broad comparison against external baselines under the same leakage-safe evaluation protocol. These factors should be considered when interpreting clinical readiness. In particular, reconstruction error can increase during both seizure windows and physiologically ambiguous non-seizure events, which limits biological specificity when that quantity is interpreted in isolation.

Future work

To address sensitivity limitations, future work will incorporate temporal sequence modeling through recurrent or attention-based architectures to capture seizure evolution across consecutive windows and improve detection of subtle ictal patterns.

To establish broader applicability, multi-center clinical validation across diverse patient populations including adults, ambulatory recordings, and varied recording conditions is essential to assess generalization beyond the pediatric hospital setting.

Alternative VAE architectures with morphology-aware latent encoders may enhance discriminative capacity and reduce performance gaps between patients with well-structured versus subtle seizure morphologies. Comparison with baseline methods and external validation on independent datasets are also recommended.

Conclusion

This study presented a patient-independent hybrid seizure-detection framework for long-term scalp EEG, in which fold-specific generative representation learning, hybrid feature construction, boosted-tree classification, and operating-point calibration were performed within a strict leave-one-patient-out design. In each fold, one patient was reserved exclusively for final testing, while channel harmonization, feature refinement, model fitting, and calibration were restricted to the remaining training patients.

Across the 24 held-out evaluation cases, corresponding to 23 distinct individuals in the public CHB-MIT release, the framework achieved mean accuracy of 0.9964, mean balanced accuracy of 0.9411, mean specificity of 0.9967, and mean false alarms per hour of 4.7154. Across the seizure-positive held-out cases, the mean sensitivity, F1-score, ROC-AUC, and average precision were 0.5505, 0.3543, 0.9496, and 0.2990, respectively. These findings show that the framework provides strong non-seizure discrimination and useful ranking performance under a strict held-out-patient evaluation protocol, while seizure sensitivity remains more moderate in the seizure-positive subset.

Within the CHB-MIT pediatric dataset, the study supports the use of hybrid latent and handcrafted seizure-sensitive features for seizure detection under heterogeneous channel availability and severe class imbalance. At the same time, the observed sensitivity indicates that further improvement is still needed for more complete seizure capture. External validation on independent datasets, evaluation on broader patient populations, and incorporation of richer temporal modeling remain important next steps for improving robustness and generalizability.

Data availability

The EEG recordings used in this study were obtained from the publicly available CHB-MIT Scalp EEG Database, hosted on PhysioNet. This dataset is openly accessible for research purposes under the PhysioNet Credentialed Health Data License. No additional datasets were generated or used in this study. All preprocessing code, feature-engineering scripts, and modelevaluation pipelines developed for this work are available from the corresponding author upon reasonable request.

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Author contributions

S.S.F. designed the study, developed the methodology, performed data processing and artifact handling, implemented the models, conducted the experiments, analyzed and validated the results, and wrote the manuscript. A.R. contributed to study design, data curation, methodology development, model implementation and validation, and manuscript writing and review. J.B. assisted with data processing, results analysis, figure preparation, and validation of the manuscript. S.P. contributed to technical discussions, result interpretation, validation, manuscript revision, and writing support. H.L. provided conceptual guidance, supervised the research, supported the study design, secured financial and technical resources, reviewed the manuscript, and finalized the submission.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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